Numerical equations

1 What is an equation?

2 State whether the following equations are true or false:

a $22 + 6 = 4 \times 7$

b $51 - 6 = 18 + 4 \times 8$

c $5 \times 8 - 27 = 19$

d $\frac{96}{8} = 106 - 94$

e $8 + 16 = 6 \times 4$

f $57 - 4 = 15 + 6 \times 7$

g $5 \times 7 - 19 = 13$

h $\frac{36}{6} = 52 - 46$

i $55 - 24 = 7 \times 3$

j $32 + 8 = 20 + 4 \times 6$

k $3 \times 6 + 7 = 32 - 8$

l $\frac{28}{4} + 7 = 30 - 16$

Equations involving unknown values

3 Fill in the blank to make the following equations true:

a $4 + \square = 8$

b $3 \times \square = 30$

c $8 \times \square = 35 - 3$

d $\square \div 3 = 9 - 6$

e $8 + \square = 17$

f $8 \times \square = 88$

g $4 \times \square = 27 - 3$

h $\square \div 6 = 9 - 3$

i $\square + 18 = 35$

j $\square \times 11 = 77$

k $\square \times 8 = 42 - 10$

l $36 \div \square = 15 - 6$

4 Consider the equation $3x + 8 = 23$.

a Find the value of each expression:

i $3 \times 1 + 8$ **ii** $3 \times 2 + 8$ **iii** $3 \times 7 + 8$ **iv** $3 \times 6 + 8$

v $3 \times 5 + 8$

b Use your answers from part (a) to solve the equation $3x + 8 = 23$.

5 Find the value of the variable that will make the following equations true:

a $x + 18 = 30$ **b** $8x = 96$ **c** $28 - x = 21$ **d** $\frac{x}{9} = 3$

e $12 + m = 25$ **f** $s - 7 = 14$ **g** $11u = 66$ **h** $\frac{m}{12} = 4$

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Word problems

6 For each of the following word sentences, write a matching equation:

a The quotient of g and seven is equal to eleven.

b The product of seven and y is equal to x less than twenty five.

c The product of eight and s is taken away from sixty six. The result is equal to ten.

d Sixteen less than m is equal to twenty six.

e Twelve less than q is equal to twenty five.

f The product of twelve and r is taken away from seventy two. The result is equal to thirty six.

g The quotient of p and ten is equal to seven.

h The product of five and q is equal to p less than twenty nine.

7 For each of the following equations, write a matching word sentence:

a $8x = 32$

b $\frac{63}{x} = 9$

c $6x + 4 = 60 - 20$

d $5y = 60$

e $\frac{72}{p} = 8$

f $8x + 7 = 72 - 25$

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